Application Note

UTA 1024-MUX Adapter

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Summary

This document describes the intended use of the Vantage UTA 1024-MUX adapter, the required configuration, programming model, assembly, and the basic sequence needed to complete a 1024 element acquisition made up of multiple transmit receive events.

Table of Contents

1. Setup

Required Hardware

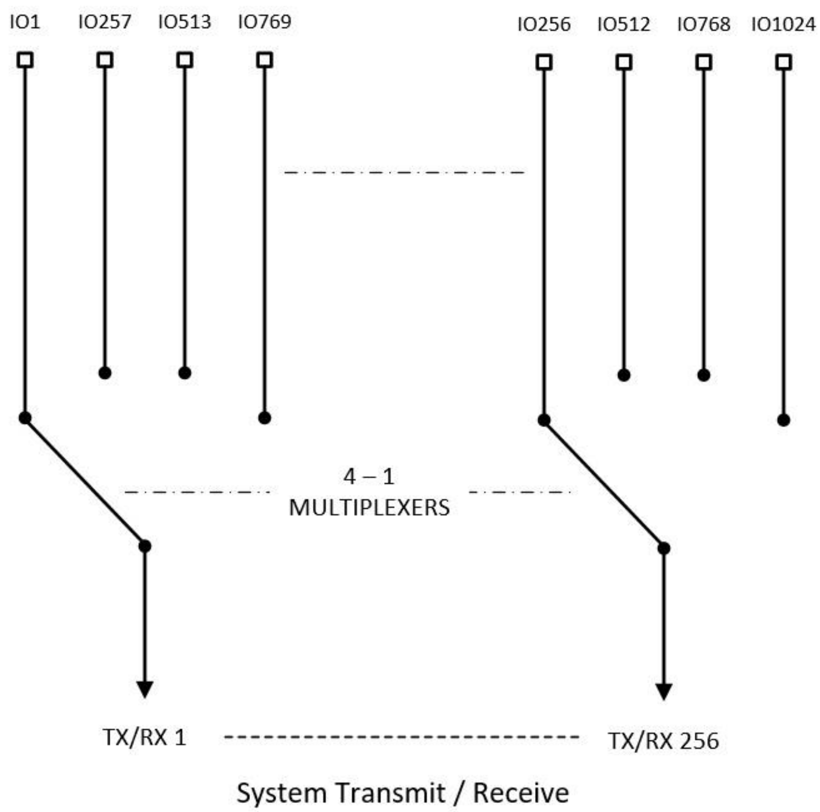
- Vantage 256
- UTA 1024-MUX Adapter

Required Software

- Vantage Release 3.2.2 or later

2. Theory of Operation

The 1024-MUX adapter supports 4-to-1 multiplexing of 256 system channels to provide connection up to 1024 elements. The user can define custom apertures using the aperture helper function within the limitations of mux exclusivity. For example, system channel 1 can be connected to elements 1, 257, 513, or 769. The system channel cannot connect to more than one mux position at a tie. See diagram below.



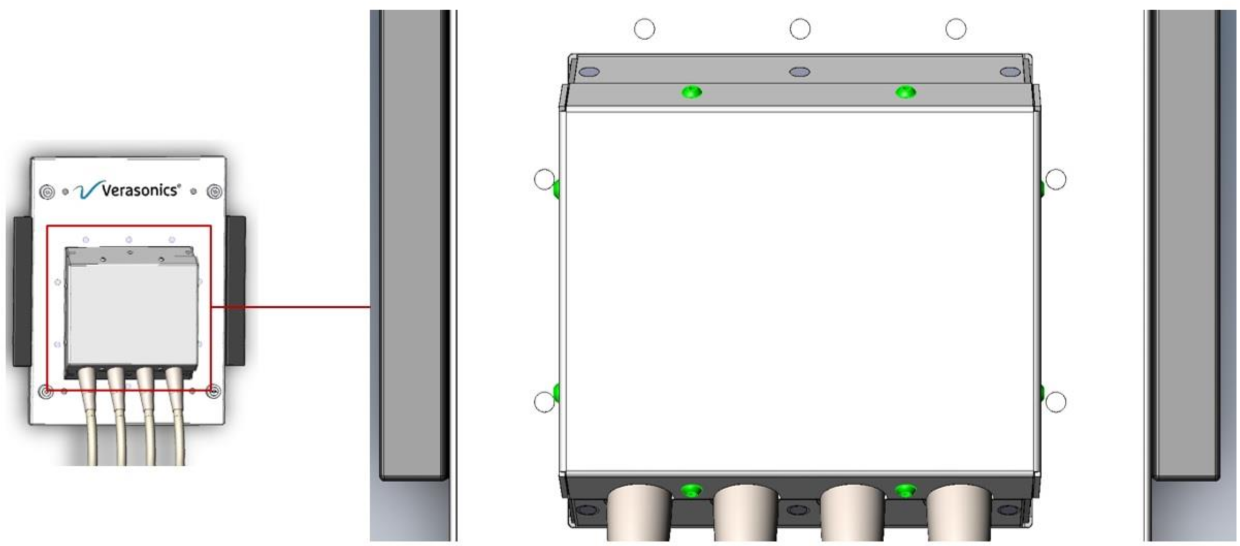
3. Assembly and Mechanical

UTA best practices

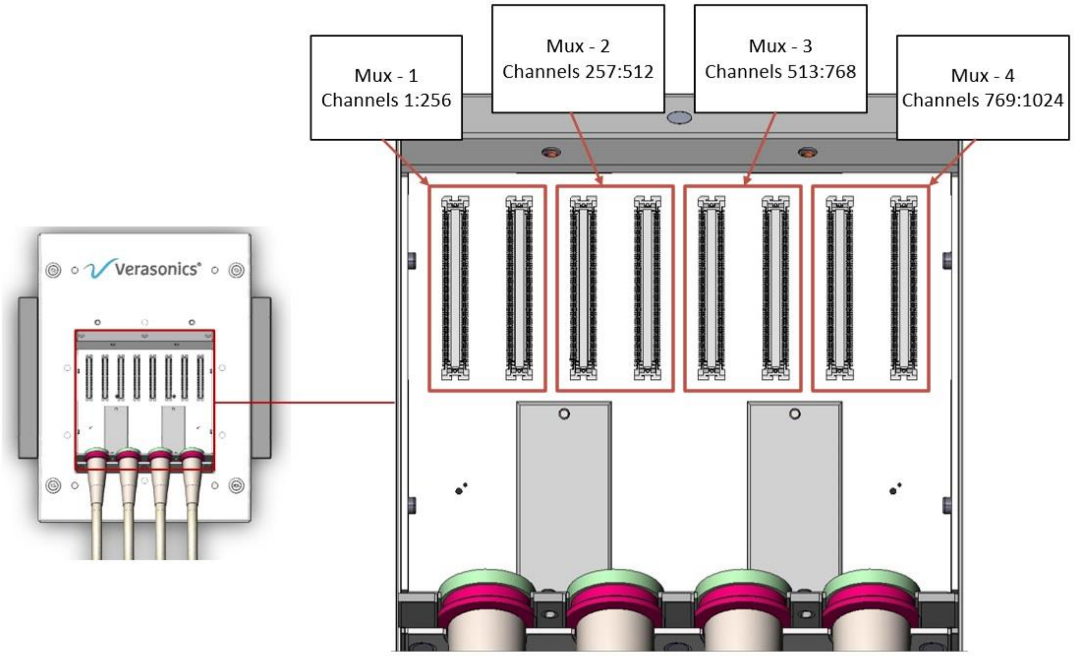
* 1024-MUX UTA assemblies should be handled with Electro Static Discharge (ESD) safe practices to avoid electrical component damage
* Transducers should ONLY be assembled when the 1024-MUX UTA is unplugged from the Vantage system
* Extreme caution should be used when plugging and unplugging termination boards so as not to damage the soldered micro coax connections
* Plugging and unplugging of termination should be minimized to preserve connector integrity/longevity

Cover removal/replacement

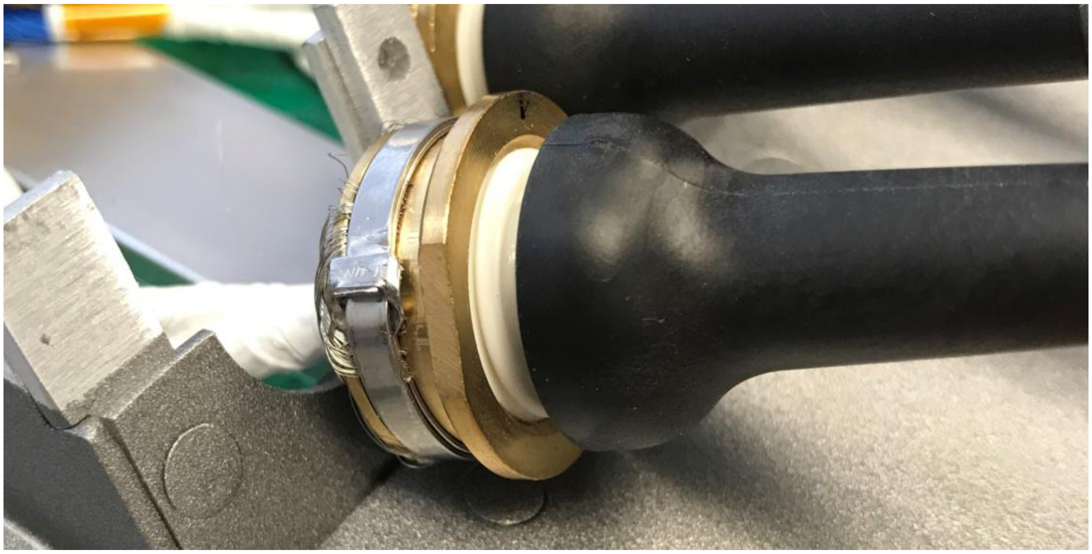
To remove or replace lid locate the 8 M3x8mm screws around the perimeter of the metal lid

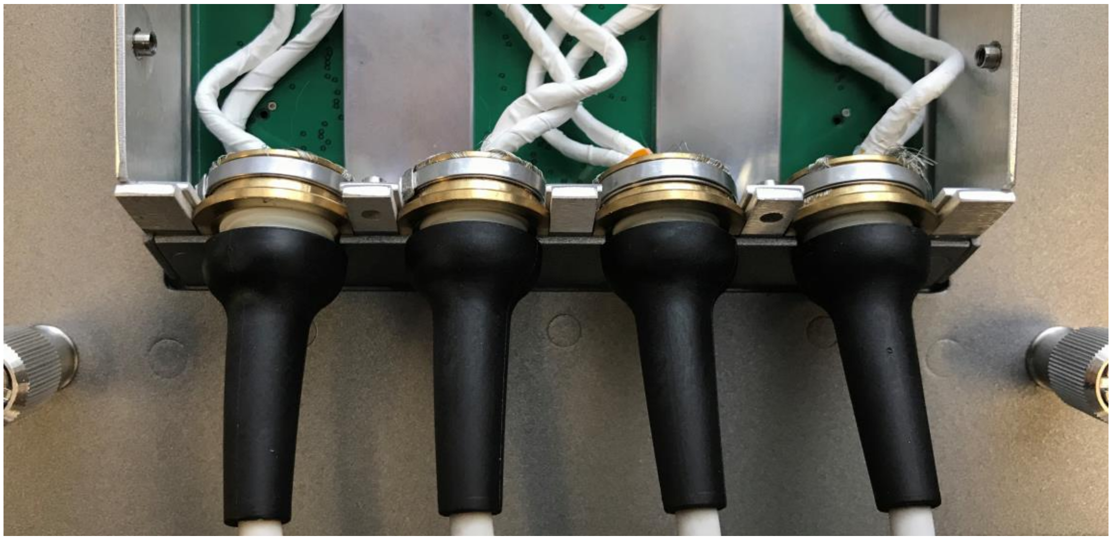


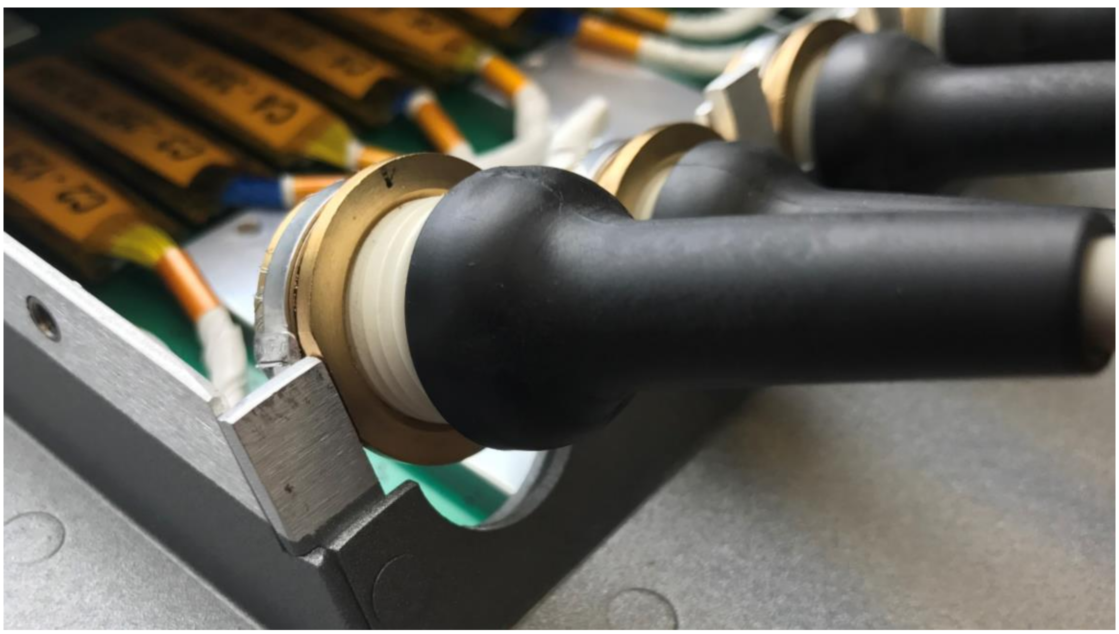
Termination board connectors



Physical assembly images







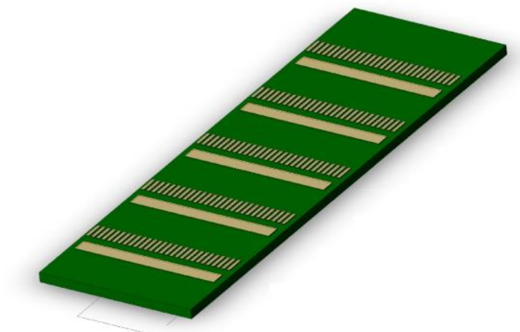
4. Software

computeUTAMux1024.m

The computeUTAMux1024.m helper function requires an aperture or an array of apertures to populate the appropriate channel-mapping and mux programming structures. To create one or more apertures a user must create a table of ones and zeros 1024 columns wide. computeUTAMux1024.m will check the input for correctness and will error out and notify the user if incorrect. If the transducer does not directly map from element to channel, the user must define a remap table for the adapter/system to use. Additional details can be found in the computeUTAMux1024.m function in the Utilities folder.

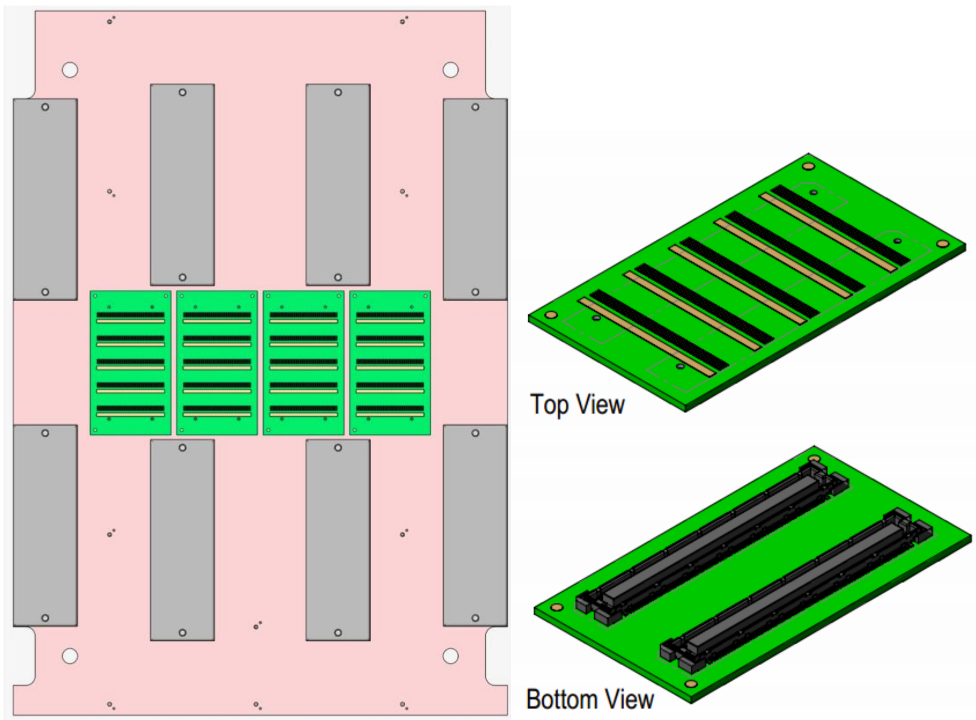
5. Vermon Cable Termination Board

Vermon 1024 element transducers are supplied with 8 termination boards compatible with the Vantage1024-MUX adapter. Each board connects to 128 elements. The default Vermon element to channel mapping is a direct map.

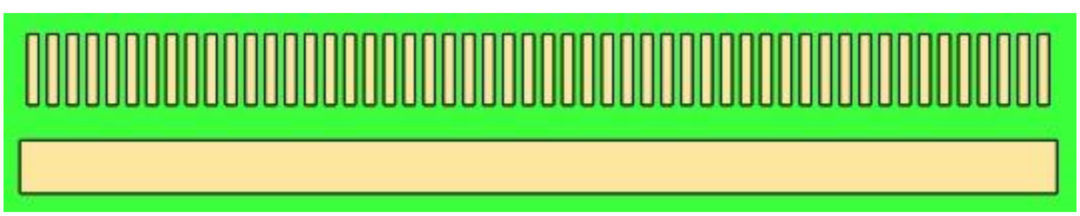


6. Optional Verasonics Cable Termination Board

Conceptual Implementation



Micro Coax Termination Pads



Pad order from left to right is 1 to 52. There are five sets of landing pads from top of the board towards the bottom which are listed as MCLP1 thru 5 below (Multi-Coax-Land-Pattern).

MCLP1

1	EL_13	14	EL_65	27	EL_141	40	EL_193
2	EL_12	15	EL_66	28	EL_140	41	EL_194
3	EL_11	16	EL_67	29	EL_139	42	EL_195
4	EL_10	17	EL_68	30	EL_138	43	EL_196
5	EL_9	18	EL_69	31	EL_137	44	EL_197
6	EL_8	19	EL_70	32	EL_136	45	EL_198
7	EL_7	20	EL_71	33	EL_135	46	EL_199
8	EL_6	21	EL_72	34	EL_134	47	EL_200
9	EL_5	22	EL_73	35	EL_133	48	EL_201
10	EL_4	23	EL_74	36	EL_132	49	EL_202
11	EL_3	24	EL_75	37	EL_131	50	EL_203
12	EL_2	25	EL_76	38	EL_130	51	EL_204
13	EL_1	26	EL_77	39	EL_129	52	EL_205

MCLP2

1	EL_26	14	EL_78	27	EL_154	40	EL_206
2	EL_25	15	EL_79	28	EL_153	41	EL_207
3	EL_24	16	EL_80	29	EL_152	42	EL_208
4	EL_23	17	EL_81	30	EL_151	43	EL_209
5	EL_22	18	EL_82	31	EL_150	44	EL_210
6	EL_21	19	EL_83	32	EL_149	45	EL_211
7	EL_20	20	EL_84	33	EL_148	46	EL_212
8	EL_19	21	EL_85	34	EL_147	47	EL_213
9	EL_18	22	EL_86	35	EL_146	48	EL_214
10	EL_17	23	EL_87	36	EL_145	49	EL_215
11	EL_16	24	EL_88	37	EL_144	50	EL_216
12	EL_15	25	EL_89	38	EL_143	51	EL_217
13	EL_14	26	EL_90	39	EL_142	52	EL_218

MCLP3

1	EL_39	14	EL_91	27	EL_167	40	EL_219
2	EL_38	15	EL_92	28	EL_166	41	EL_220
3	EL_37	16	EL_93	29	EL_165	42	EL_221
4	EL_36	17	EL_94	30	EL_164	43	EL_222
5	EL_35	18	EL_95	31	EL_163	44	EL_223
6	EL_34	19	EL_96	32	EL_162	45	EL_224
7	EL_33	20	EL_97	33	EL_161	46	EL_225
8	EL_32	21	EL_98	34	EL_160	47	EL_226
9	EL_31	22	EL_99	35	EL_159	48	EL_227
10	EL_30	23	EL_100	36	EL_158	49	EL_228
11	EL_29	24	EL_101	37	EL_157	50	EL_229
12	EL_28	25	EL_102	38	EL_156	51	EL_230
13	EL_27	26	EL_103	39	EL_155	52	EL_231

MCLP4

1	EL_52	14	EL_104	27	EL_180	40	EL_232
2	EL_51	15	EL_105	28	EL_179	41	EL_233
3	EL_50	16	EL_106	29	EL_178	42	EL_234
4	EL_49	17	EL_107	30	EL_177	43	EL_235
5	EL_48	18	EL_108	31	EL_176	44	EL_236
6	EL_47	19	EL_109	32	EL_175	45	EL_237
7	EL_46	20	EL_110	33	EL_174	46	EL_238
8	EL_45	21	EL_111	34	EL_173	47	EL_239
9	EL_44	22	EL_112	35	EL_172	48	EL_240
10	EL_43	23	EL_113	36	EL_171	49	EL_241
11	EL_42	24	EL_114	37	EL_170	50	EL_242
12	EL_41	25	EL_115	38	EL_169	51	EL_243
13	EL_40	26	EL_116	39	EL_168	52	EL_244

MCLP5

1	N/C	14	EL_53	27	EL_192	40	EL_246
2	N/C	15	EL_117	28	EL_191	41	EL_247
3	EL_64	16	EL_118	29	EL_190	42	EL_248
4	EL_63	17	EL_119	30	EL_189	43	EL_249
5	EL_62	18	EL_120	31	EL_188	44	EL_250
6	EL_61	19	EL_121	32	EL_187	45	EL_251
7	EL_60	20	EL_122	33	EL_186	46	EL_252
8	EL_59	21	EL_123	34	EL_185	47	EL_253
9	EL_58	22	EL_124	35	EL_184	48	EL_254
10	EL_57	23	EL_125	36	EL_183	49	EL_255
11	EL_56	24	EL_126	37	EL_182	50	EL_256
12	EL_55	25	EL_127	38	EL_181	51	N/C
13	EL_54	26	EL_128	39	EL_245	52	N/C

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